Berk Çiçek

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EDUCATION

Bilkent University, Ankara, Turkey — 2022 – 2025
M.Sc. in Computer Engineering — *CGPA: 3.96/4.0*
Thesis: *Contact-VLA: Zero-Shot Planning and Control for Contact-rich Manipulation*
Advisor: Asst. Prof. Ozgur Oguz

Bilkent University, Ankara, Turkey — 2017 – 2022
B.S. in Mechanical Engineering — *CGPA: 3.33/4.0*

RESEARCH EXPERIENCE

Graduate Research Assistant — Sep 2022 – Present
LIRA Lab, Bilkent University, Ankara, Turkey
Advisor: Asst. Prof. Ozgur Oguz

- Developing zero-shot manipulation frameworks combining Vision-Language models with motion planners for contact-rich robotic manipulation
- Designed and implemented **CoRAL** (Contact-Rich Adaptive LLM-based Control), a neuro-symbolic framework integrating LLMs with MPPI controllers for adaptive physical reasoning (accepted at RSS 2026)
- Built end-to-end pipelines for robotic manipulation using **Franka Research 3** robot, including perception (FoundationPose), planning (MPPI), and reactive control modules
- Developed hybrid sequential manipulation planner (**H-MaP**) published in RA-L 2025, enabling complex multi-stage manipulation tasks
- Implemented vision transformer-based grasping system (**FViT-Grasp**) achieving real-time inference for 6-DoF grasp detection

Visiting Researcher — June 2021 – Aug 2021
National Research Center of Italy (CNR-IPCF), Messina, Italy
Supervisor: Dr. Onofrio Maragò

- Developed computational models for Janus particles under optical potentials and thermophoresis using MATLAB
- Contributed to research published in *ACS Photonics* on optically driven micro-engines with full orbital motion control

Undergraduate Researcher — Mar 2020 – June 2022
Bilkent University, Ankara, Turkey
Supervisor: Asst. Prof. Luca Biancofiore

- Conducted computational fluid dynamics (CFD) simulations and particle modeling using LAMMPS, OpenFOAM, and MATLAB
- Investigated self-driven complex particles under optical potentials for micro-robotics applications

PUBLICATIONS

Selected Publications

- **CoRAL: Contact-Rich Adaptive LLM-based Control for Robotic Manipulation**
  Accepted at *Robotics: Science and Systems (RSS)*, 2026.

Journal Publications

- **H-MaP: Iterative and Hybrid Sequential Manipulation Planner**
  IEEE Robotics and Automation Letters (RA-L), 2025. To be presented at ICRA 2026.
- **Interpretable Responsibility Sharing as a Heuristic for Task and Motion Planning**
  Robotics and Autonomous Systems 201, 105436
- **Optically Driven Janus Micro Engine with Full Orbital Motion Control**
  ACS Photonics, 2023.

Preprints & Under Review

- **FViT-Grasp: Grasping Objects With Using Fast Vision Transformers**
  arXiv:2311.13986.

RESEARCH INTERESTS & TECHNICAL SKILLS

Research Interests:
Robotic manipulation, Vision-Language-Action (VLA) models, Foundation models for robotics (LLMs/VLMs), sampling & optimization-based motion planning, Model Predictive Control, Neuro-symbolic reasoning, Zero-shot task generalization, Embodied AI

Programming & Tools: Python, C++, MATLAB, Git, Linux
Robotics Platforms: Franka Emika Panda, Gazebo, MuJoCo, ROS/ROS2, Drake, RAI, RoboSuite
Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, Weights & Biases
Computer Vision: FoundationPose, Vision Transformers (ViT), OpenCV, 6-DoF pose estimation
Simulation & Modeling: OpenFOAM, LAMMPS, Ansys, Comsol
Data Engineering: Spark, SQL, Airflow, Tableau
CAD & Design: CATIA, SolidWorks, Unity
Languages: Turkish (Native), English (Fluent), Italian (Beginner), Spanish (Beginner), German (Beginner)

TEACHING EXPERIENCE

Teaching Assistant, Bilkent University

- **CS-549: Learning for Robotics** Fall 2023 -Spring 2024
- **CS-202: Fundamental Structures of Computer Science II** Spring 2024
- **CS-115: Introduction to Programming with Python** Fall 2022 – Spring 2023

AWARDS & HONORS

- **Trendyol AI Hackathon Winner** – First place with presented AI-powered search algorithm 2025
- **TAI-TUSAS Hangar Award** – 90,000 TL prize for Smart Road Maintenance Platform 2023
- **Dincer Logistics Hackathon Winner** – 100,000 TL prize for WareMan computer vision warehouse system 2023
- **European Union EIT Jumpstarter** – Selected for Bilabel data labeling platform 2023
- **Bilkentpreneurs Competition** – 2nd place for Bilabel startup 2024
- **Odgers Berndtson CEOx1Day Winner** – Future leadership development program 2023

SELECTED PROJECTS

- **PuzzleWorld-3D** (Ongoing) – Benchmark platform for testing reasoning capabilities of embodied AI and foundation model agents via API-based 3D manipulation tasks. puzzleworld-3d.com 2026
- **Bilabel (Co-Founder)** – Gamification-based data labeling and validation platform for computer vision applications 2023–2024
- **Hand Pose Estimation with LTF** – CS554 Computer Vision course project 2023
- **Grasp Point Detection with F-ViT** – CS549 Learning for Robotics course project 2023
- **Multi-Directional Power Soccer Chair** – Graduation project: wheelchair with VR game interface using Unity and Photon 2022

INDUSTRY EXPERIENCE

Data Scientist, Trendyol, Istanbul, Turkey Feb 2025 – Present
Developing AI-powered search algorithms for Turkey's largest e-commerce platform

Data Scientist, Invent Analytics, Istanbul, Turkey July 2024 – Jan 2025
Developed AI-driven predictive models for demand forecasting and inventory optimization in retail sector

REFERENCES

- **Dr. Ozgur Oguz**, Assistant Professor, Bilkent University
  ozgur.oguz@bilkent.edu.tr
- **Dr. Onofrio Maragò**, Research Director, CNR-IPCF, Italy
  onofrio.marago@cnr.it
- **Dr. Luca Biancofiore**, Associate Professor, University of L'Aquila, Italy
  luca@bilkent.edu.tr